Chap 15

15.2 Area, Moments & Center of mass

Area : $\boxed{A = \iint_R dA}$

Average value of f over R : $\boxed{\dfrac{1}{\text{area of } R} \iint_R f \, dA}$

Mass: $\boxed{M = \iint_R \delta(x,y) \, dA}$ $\delta(x,y)$ is the density at (x,y)

First moments : $\boxed{\begin{array}{l} M_x = \iint_R y\,\delta(x,y)\, dA, \\ M_y = \iint_R x\,\delta(x,y)\, dA \end{array}}$

Center of mass: $\boxed{\bar{x} = \dfrac{M_y}{M}, \quad \bar{y} = \dfrac{M_x}{M}}$

Moments of Inertia (second moments):

About x-axis $\boxed{I_x = \iint y^2 \delta(x,y)\,dA}$

About y-axis $\boxed{I_y = \iint x^2 \delta(x,y)\,dA}$

About a line L $\boxed{I_L = \iint r^2(x,y)\delta(x,y)\,dA}$

where (x,y) = distance from (x,y) to L

About the origin (polar moment): $\boxed{I_0 = \iint (x^2+y^2)\delta(x,y)\,dA = I_x + I_y}$

Radii of gyration:

About x-axis $R_x = \sqrt{I_x/M}$

About y-axis $R_y = \sqrt{I_y/M}$

About z-axis $R_z = \sqrt{I_0/M}$

15.3 Double Integrals in Polar Form

Area in Polar Coordinates : $\boxed{A = \iint_R r \, dr \, d\theta}$

Cartesian Integrals → Polar Integrals :

$$\iint_R f(x,y) \, dx \, dy = \boxed{\iint_G f(r\cos\theta, r\sin\theta) \, r \, dr \, d\theta}$$

15.4 Triple Integrals in Rectangular Coordinates

Volume: $\boxed{V = \iiint_D dV \ , \ \iiint_D dz \, dy \, dx}$

Average value of F over D

$$= \boxed{\frac{1}{\text{Volume of } D} \iiint_D F \, dV}$$

15.5 Masses and Moments in Three Dimensions

Mass $M = \iiint_D \delta \, dV$ $\quad (\delta = \delta(x, y, z) = \text{density})$

First moments about the coordinate planes:

$$M_{yz} = \iiint_D x \delta \, dV, \quad M_{xz} = \iiint_D y \delta \, dV, \quad M_{xy} = \iiint_D z \delta \, dV$$

Center of mass:

$$\bar{x} = \frac{M_{yz}}{M}, \quad \bar{y} = \frac{M_{xz}}{M}, \quad \bar{z} = \frac{M_{xy}}{M}$$

Moments of Inertia:

$$I_x = \iiint (y^2 + z^2) \delta \, dV$$

$$I_y = \iiint (x^2 + z^2) \delta \, dV$$

$$I_z = \iiint (x^2 + y^2) \delta \, dV$$

Moments of inertia about line L:

$$I_L = \iiint r^2 \delta \, dV$$

15.6 Triple Integrals in Cylindrical & Spherical Coordinates.

Equations relating rectangular (x, y, z) & cylindrical (r, θ, z) coordinates

$$\boxed{\begin{array}{l} x = r\cos\theta, \quad y = r\sin\theta, \quad z = z, \\ r^2 = x^2 + y^2, \quad \tan\theta = y/x \end{array}}$$

Equations relating spherical coordinates to Cartesian & cylindrical coordinates

$$\boxed{\begin{array}{l} r = \rho\sin\phi, \quad x = r\cos\theta = \rho\sin\phi\cos\theta, \\ z = \rho\cos\phi, \quad y = r\sin\theta = \rho\sin\phi\sin\theta, \\ \rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2} \end{array}}$$

Integrate in Spherical Coordinates:

$$\iiint_D f(\rho, \phi, \theta)\, dV$$

Volume in Spherical Coordinates:

$$\boxed{V = \iiint_D \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta}$$

$$dV = dx\, dy\, dz$$
$$= dz\, r\, dr\, d\theta$$
$$= \rho^2 \sin\phi \, d\rho\, d\phi\, d\theta$$

15.7 Substitutions in Multiple Integrals

Jacobian of the coordinate transformation

$x = g(u,v)$, $y = h(u,v)$:

$$J(u,v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\frac{\partial x}{\partial v}$$